INCH-POUND

MIL-R-874/2

21 June 1990

MILITARY SPECIFICATION

RESISTOR NETWORK, FIXED, FILM, ESTABLISHED RELIABILITY, STYLE RZR20, 16 PIN DIP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RZR10, 16 pin dual in-line package (DIP) resistor network. This style is available in all characteristics and resistance tolerances.

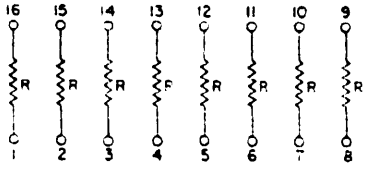
1.2 Part or Identifying Number (PIN). Resistor networks covered by this specification shall be identified by a PIN which shall consist of the basic number of this specification and a coded number. The PIN shall be in the following form:

M874 — Military specification number

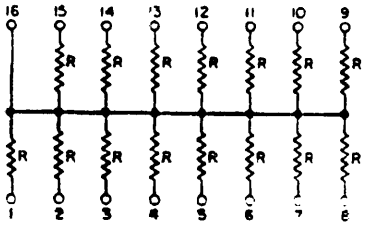
H02F1002AS — Coded number

The coded number shall be derived in accordance with MIL-R-874.

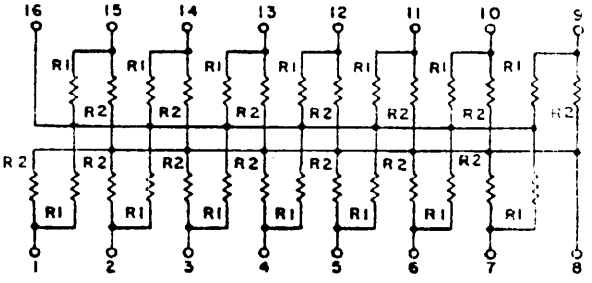
1.2.1 Schematics. The schematic of the resistor network shall be identified by a single letter in accordance with the drawings below. The resistor element R_{ref} shall be the reference resistor element used in determining the ratio accuracy (when applicable).



SCHEMATIC "A"



SCHEMATIC "B"



SCHEMATIC "J"

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5905

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1.2.1.1 Standard resistance values. The standard resistance values and the resistance value designators for the "J" schematic are as specified in table I.

TABLE I. Standard resistance values.

Resistance designator	R1 (ohms)	R2 (ohms)	Resistance designator	R1 (ohms)	R2 (ohms)
A001	82	130	A010	330	470
A002	120	200	A011	330	680
A003	130	210	A012	1.5k	3.3k
A004	160	260	A013	3.0k	6.2k
A005	180	240	A014	180	270
A006	180	390	A015	270	270
A007	220	270	A016	560	560
A008	220	330	A017	560	1.2k
A009	330	390	A018	620	2.7k

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-R-874 - Resistor Networks, Fixed, Film, Established Reliability, General Specification for.
MIL-R-39032 - Resistors, Packaging of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Aveue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

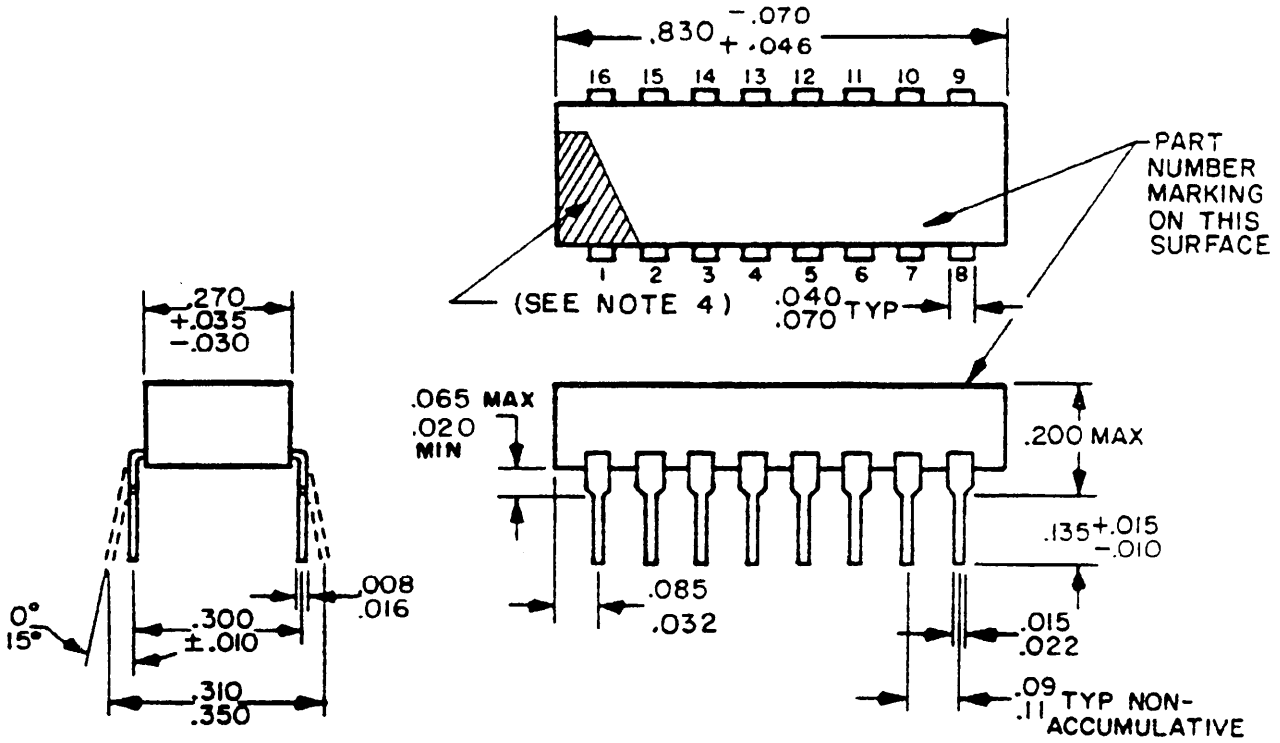
3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-874, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be as specified in table II, based on a full load operation at an ambient temperature of 70°C. For operation at temperatures higher than 70°C, derate in accordance with MIL-R-874.

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Inches	mm	Inches	mm
.008	0.20	.070	1.78
.010	0.25	.085	2.16
.015	0.38	.090	2.29
.016	0.41	.110	2.79
.020	0.51	.135	3.43
.022	0.56	.200	5.08
.030	0.76	.270	6.86
.032	0.81	.300	7.62
.035	0.89	.310	7.87
.040	1.02	.350	8.89
.046	1.17	.830	21.08
.065	1.65		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The picturization of the styles above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.
4. Pin 1 locator shall be a dot, notch, stripe, or numeral 1 adjacent to pin number 1, in the shaded area.

FIGURE 1. RZR20 style resistors.

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TABLE II. Power ratings.

Schematic	Characteristic	Element power rating (watts)	Network power rating (watts)
A	H	0.2	1.6
	K	0.2	1.6
	M	0.2	1.6
B	H	0.1	1.5
	K	0.1	1.5
	M	0.1	1.5
J	H	0.050	1.4
	K	0.050	1.4
	M	0.050	1.4

3.4 Power conditioning. The power applied for power conditioning for schematics "A", "B", and "J" shall be 1.5 times rated power.

3.5 Voltage rating. The maximum continuous working voltage for each style shall not exceed 100 volts dc or ac rms.

3.6 Resistance. Minimum and maximum resistance values shall be as specified in table III.

TABLE III. Resistance values.

Resistance tolerance	Characteristic	Minimum resistance (ohms)	Maximum resistance (megohm)
B (±0.1 percent)	H	100	1
D (±0.5 percent)	H, K	100	1
F (±1.0 percent)	H, K, M	10.0	1
G (±2.0 percent)	K, M	10.0	1
J (±5.0 percent)	K, M	10.0	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-874.

4.2 "J" schematic tests. For the tests on "J" schematic, the dc resistance measurements, resistance temperature characteristic measurement, and short time overload power application shall be made between each terminal and the applicable common terminal without compensation for the shunt circuits. The pin to pin dc resistance measurements shall be used directly to determine the acceptability to section 3 requirements.

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.022 0.56 11 .200 5.08

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-874/2	Resistor Network, Fixed, Film, Established Reliability, Style RZR20, 16 Pin Dip

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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